SC3: *A Practical Introduction to Physics-informed Deep Learning for Surrogate Models of Drinking Water Systems*

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Your Instructors

- Machine Learning researchers from Bielefeld University, Germany
    - Part of the ERC Water Futures project

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Outline of Today

Time	Topics
14:00	Introduction & Welcome
	Surogate models vs. digital twins
	Types & Implementations of surrogates
14:15	Recap of Machine Learning (ML) and Deep Learning
	Basic concepts, evaluation, hyper-parameter tuning, etc.
	Surrogate models: Challenges for ML
	Data-driven mitigation strategies (i.e., scenario generation)
14:45	Physics-informed/inspired Methods -- Part I
	Intro to Physics-informed Neural Networks (PINNs)
	Hands-On: How to build and train a PINN in Python
15:30	Coffee Break
16:00	Physics-informed/inspired Methods -- Part II
	Operators vs. PINNs
16:25	Case study: SOTA Hydraulic Surrogate Model
	Hydraulic surrogates and their applications
	Analyzing the problem and identifiying relevant quantities and laws
	A SOTA Hydraulic Surrogate
	Hands-On: Usage and Evaluation of the Surrogate in Python
17:15	Closing, Summary, and Q&A
17:30	END

We are here!

TQRCG

Course material @GitHub
https://tinyurl.com/2m4vy8vp

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Introduction to Surrogate Models

- Surrogates vs. Simulators
  - Surrogates: fast approximation of simulators
- Simulators:
  - Accurate
  - Computational inefficient
  - No back-propagation (e.g., not differentiable)
  - Can not deal with uncertainties
- Surrogates:
  - Computational more efficient
  - If differentiable, "easy" backpropagation
  - Useful for downstream tasks where sensitivity analysis is required
  - Uncertainties could be handled – but needs an extension
  - Less accurate

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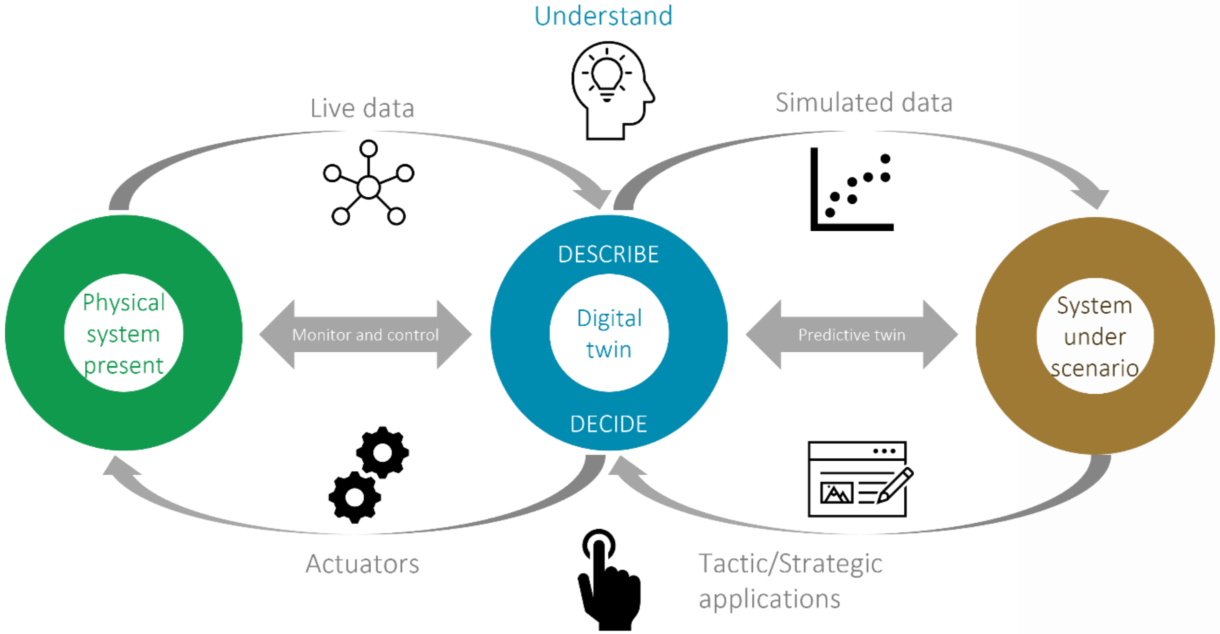
Digital Twins

- *Digital Twins:* real-time virtual clone of a real system
  - Uses real-time data (e.g., sensor measurements)
  - Surrogates are often part of digital twins for speeding up simulations
    - e.g. What-If analysis, planning, etc.



https://www.kwrwater.nl/en/projecten/digital-twins-for-the-water-sector/

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Digital Twins for Water Distribution Systems

- *Calibrate* models with (real-time) sensor readings
- Hydraulic, and quality *surrogates*
    - Approximate or improve EPANET and EPANET-MSX models
    - Handle uncertainties
    - ....

=> But how to implement a surrogate?

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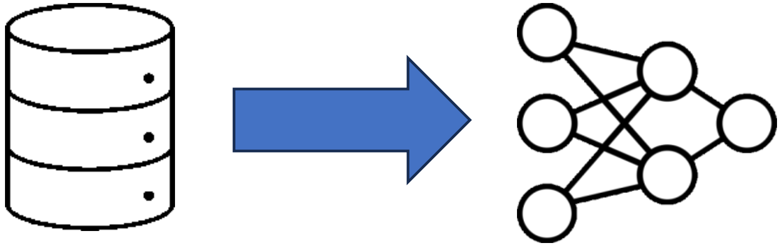
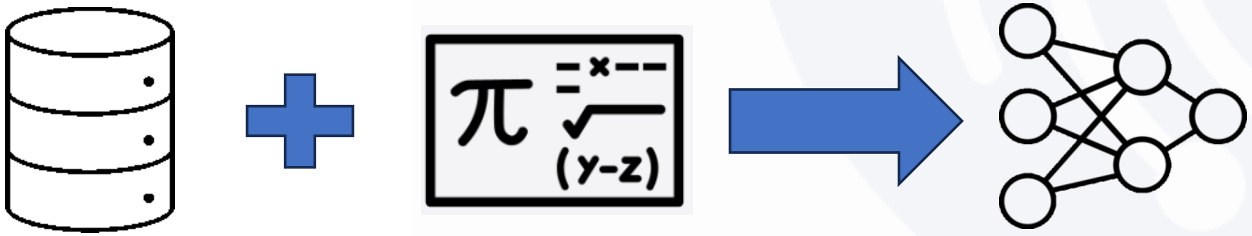
Implementations of Surrogate Models

Data-driven vs. Physics-informed

- All physics are *implicitly* encoded in the training data
- "Easy" to do
- No (formal) guarantees



- *Explicit* integration of physics
  - e.g. laws, conservations, etc.
- Training data + physics constraints
- Challenging but yields (formal) guarantees



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Next: Recap of Machine Learning